

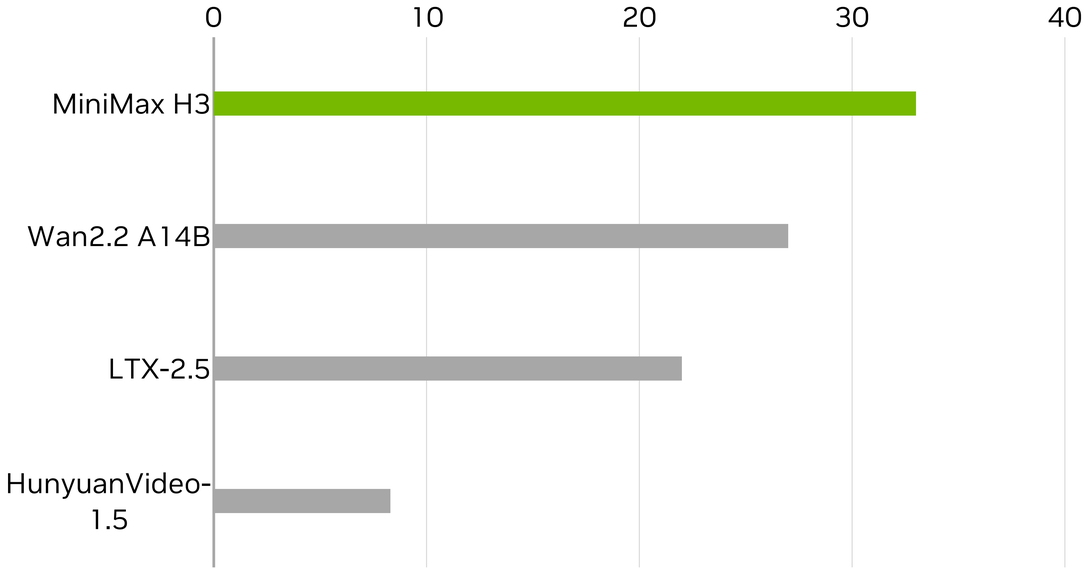
Toward Real-time Generation with MiniMax H3

Li Shunyang Devtech Compute NVIDIA

2026-08-23

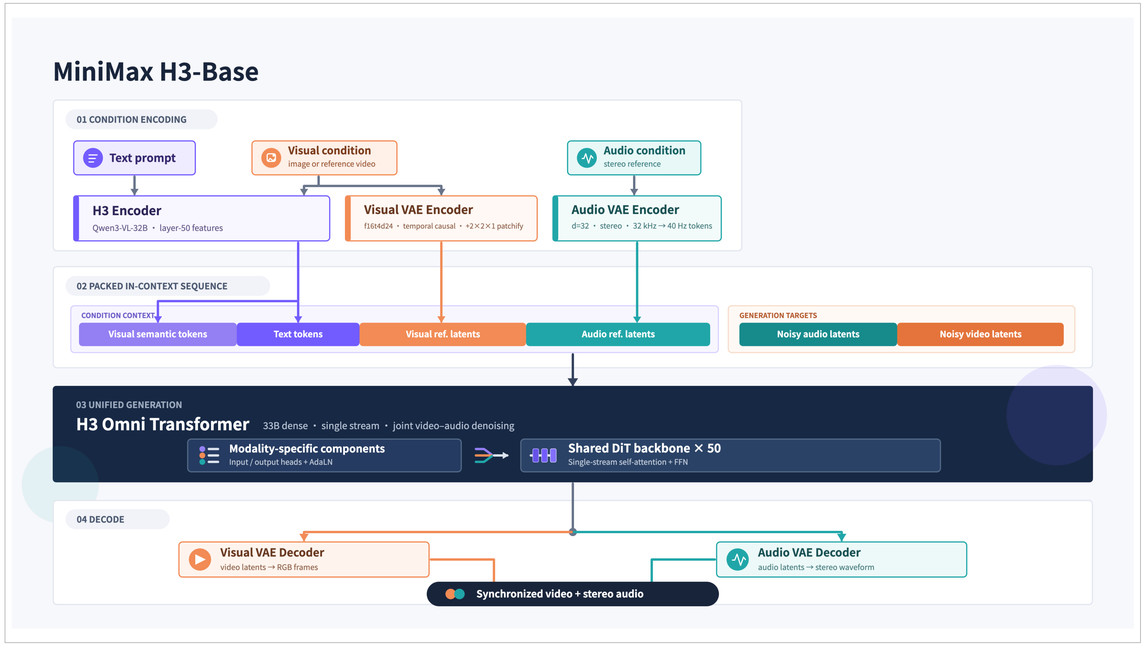
Real Time Is an End-to-End Budget

Generate about five seconds of audio and video within the same time budget.



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

The backbone is only part of the budget. Encoding, communication and audio–video output all contribute to latency.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

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SM120 Cards

GPU	SMs	Memory	Power	Interconnect
RTX PRO 5000	110	72 GB GDDR7 ECC	300 W	PCIe Gen5 ×16 No GPU NVLink
RTX PRO 6000 **Workstation / Server**	188	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	170	32 GB GDDR7	575 W	PCIe Gen5 ×16 No NVLink
DGX Spark **GB10 · SM121**	48	128 GB Unified LPDDR5X	140 W SoC	CPU–GPU NVLink-C2C 200 Gb/s ConnectX-7

×8

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Baseline: Latency Breakdown

E2E: 213.402 s · Generated video: 960 × 544, 5.167 s

Reference configuration

- 8 × PRO 5000 GPUs (SM120)
- Original Ref2VA
- 50 steps · BF16 · dense attention
- DiT TP1 / Ulysses8
- Text encoder TP8 · VAE tiles8
- Native VAE · FP16 decode

GPU work by operator

	Mean s / GPU	Share
NCCL · includes waiting	122.182	**57.21%**
Attention	47.105	**22.06%**
GEMM	39.650	**18.57%**
Norm / activation / elementwise	2.288	1.07%
Layout / device copies	1.929	0.90%
CUDA memcpy / memset	0.064	0.03%
Convolution	0.192	0.09%
Other kernels	0.143	0.07%

Nsight GPU work includes overlap and waiting; shares exclude CPU work and idle gaps.

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Optimization - Distillation

Reduce how often the model runs.

Base H3

50 model evaluations



Distilled student

8 model evaluations



84%

fewer evaluations

Less repeated model work also means fewer repeated communication rounds.

Evaluation count reduction; measured E2E results are shown in the optimization table.

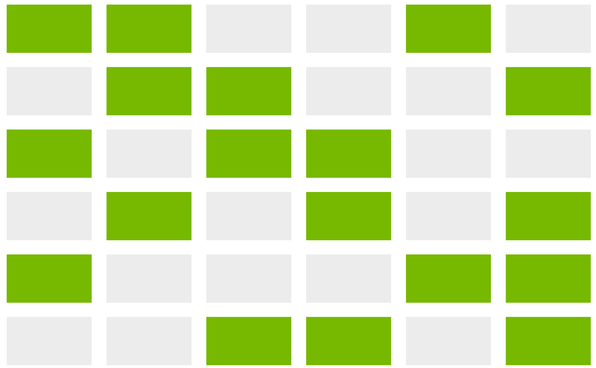
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Optimization - Video Sparse Attention + NVFP4

Choose where to compute, then lower the cost of that computation.

Video Sparse Attention

Attend to selected regions



Content-dependent selection · schematic

NVFP4

Lower precision with QAD

BF16 → NVFP4

Train for quantization error.
Keep sensitive paths at higher precision.

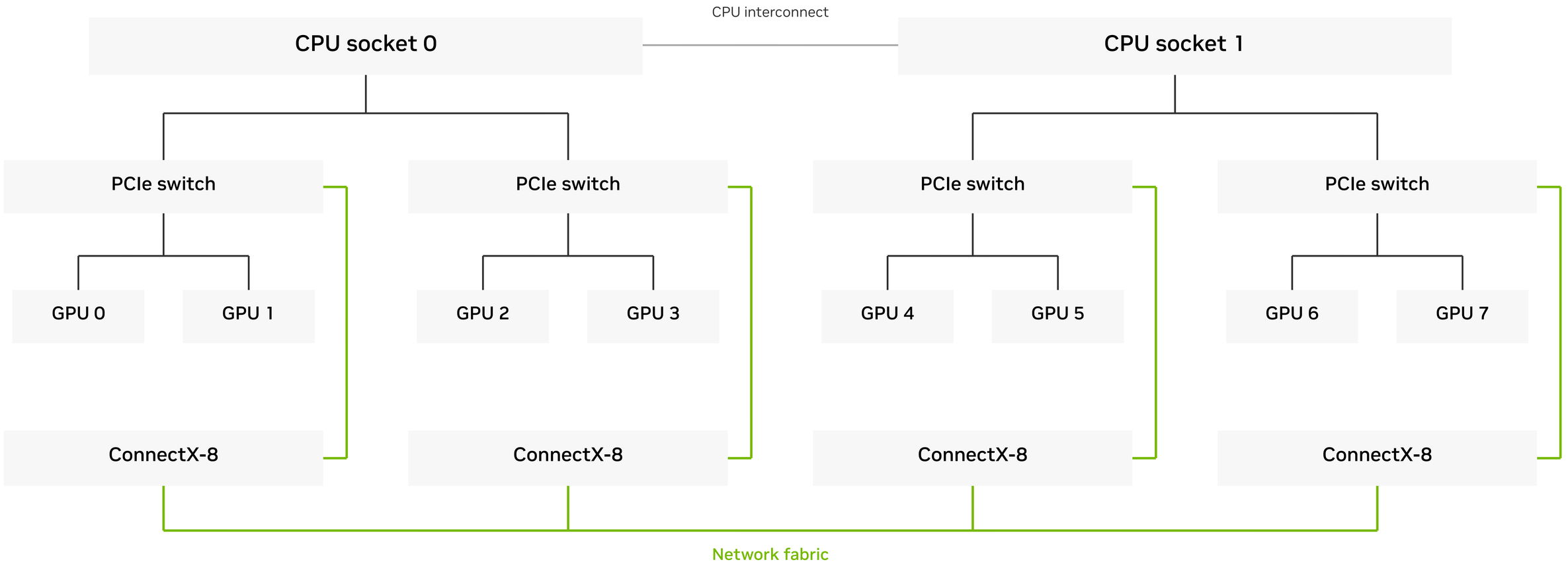
Evaluate speed and quality together on representative scenes.

Sparsity and precision changes require workload-specific quality checks.

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Optimization - Comms

Topology: the communication path matters.



Map GPU–NIC affinity before choosing the communication path.

8 × PRO 5000 · 4 × ConnectX-8 · two GPUs per PCIe subtree · no GPU NVLink

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Optimization - Comms

Choose the path, overlap the work, and reduce movement.

Route

Use GPU-affine NICs and RDMA.

Overlap

Transfer the next head band while attention runs.

Streamline

Use direct layouts and replay stable compute.

Overlap transfers with attention



Conceptual head-band schedule

3.1× faster

Recorded E2E · stages 03 → 06

Communication is a major opportunity for lossless optimization.

Grouped result: RDMA + layout + head overlap + compute graphs.

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Real Time Comes from the Whole Pipeline

960 × 544 · 5.167 s output · 8 GPUs · standalone H3

Stage	Cumulative change	Median (s)	E2E speedup vs. previous	RTF
00	Baseline · original BF16 / dense / 50 evaluations	213.402	—	41.304
01	+ PDD · 50 → 8 evaluations	44.327	4.814×†	8.579
02	+ Trained VSA · native 8-step model / runtime	25.915	1.710×	5.016
03	+ QAD NVFP4 DiT / native FP4 attention	19.460	1.332×	3.767
04	+ RDMA / direct layout / V-gate overlap	6.832	2.848×	1.322
05	+ 2 / 3 / 2 head pipeline	6.407	1.066×	1.240
06	+ Compute CUDA graphs	6.193	1.035×	1.199
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420	1.143×	1.049
08–12	**+ Output and data-movement optimizations**	**4.922**	**1.101×**	**0.953**

08–12: fused tile / AAC preparation, direct I/O packing, compact routing and Q/K/V copy removal.

E2E speedup = previous displayed median ÷ current median. 08–12 compares the final result with stage 07.

Final median: 4.922 s · recheck: 4.927 s · 6/6 measured requests below 5 s

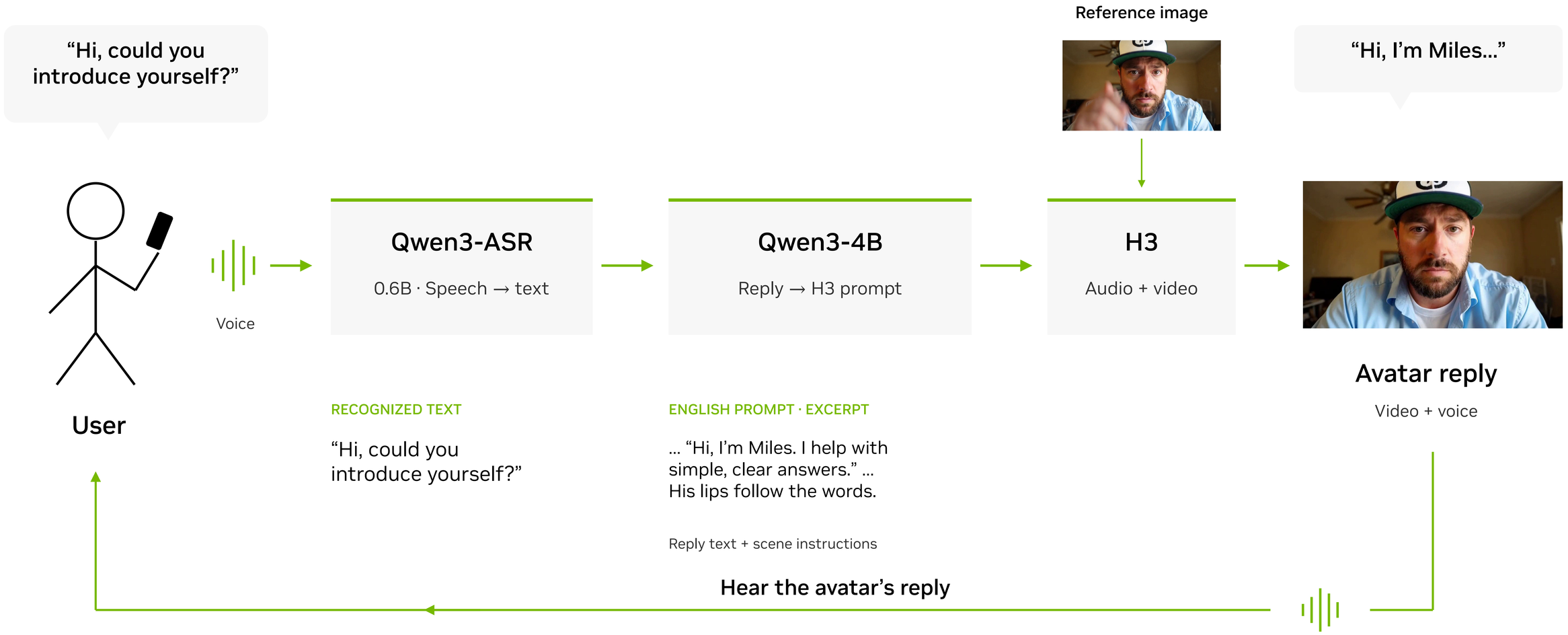
† Recorded median ratio; baseline refreshed separately (n = 2). Stages 01–12 retain n = 3 each; no isolated 50→8-step gain claimed.

RTF = latency / 5.167 s. Standalone H3; ASR / LLM excluded. Model, schedule and precision changes are not all lossless.

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Real-Time Interaction Needs the Full Loop

Speech, reasoning and audio–video generation share the response budget.



960 × 544 · ~5 s per window · 8 × RTX PRO 5000

Live Avatar: Recorded Conversation

74.55 seconds · user-recorded microphone conversation · native H3 audio + video



Play the recorded interaction · fullscreen avatar, live captions and timing overlays

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Takeaways

01 Multi-GPU optimization starts with topology

Map GPU/NIC links and locate communication bottlenecks.
Use available bandwidth efficiently, or reduce data transfer.
Communication is a key source of lossless speedups.

02 Evaluate quantization case by case

Lossy quantization behaves differently across models and workloads.
Measure quality and latency on representative cases.

03 Next: real-time response and richer interaction

Explore autoregressive diffusion for earlier responses.
Use prompt enhancers to support richer interaction.

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